Name ____________________ Math 1350, Quiz #9

1. (Section 7.3, Problem 7) Identify all of the asymptotes of $y = \frac{2x+1}{x-3}$.

$x - 3 = 0 \qquad x = 3 \quad$ VA

$y = \dfrac{2 + \frac{1}{x}}{1 - \frac{3}{x}} \rightarrow y = 2$ as $x \rightarrow \infty \qquad y = 2 \quad$ HA

2. (Section 7.4, Problem 13) Use the methods given in chapter 7 to sketch the graph of $y = \dfrac{x(x+1)}{(x-2)^2}$, i.e., specifically identify

a. any symmetries which the graph might possess,
b. any intercepts which the graph might possess,
c. any vertical asymptotes which the graph might possess,
d. any horizontal asymptotes which the graph might possess,
e. any slant asymptotes which the graph might possess,
f. intervals (on the *x*-axis) on which the sign of the graph is constant.

Then using the above information sketch the graph. Do not plot the graph point by point.

Sym None

Int. $x=0$ $(0,0)$

$x+1=0$ $(-1,0)$

VA $x=2$

HA $y=\frac{1}{1} \Rightarrow y=1$

SA None

Sign

$(-\infty,-1)$	+
$(-1,0)$	−
$(0,2)$	+
$(2,\infty)$	+